

# Notes, Solutions etc.

for George E. Martin's "Counting - The Art of Enumerative  
Combinatorics"

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# 1. Elementary Enumeration

## Combination Formula

The number of ways to choose  $r$  objects from  $n$  distinct objects is given by:

$$C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

Also written as  ${}_nC_r$  or  $\binom{n}{r}$ .

## Quickies - I

1. Addition principle:  $6 + 8 = 14$  ways.
2. Same principle, although different fruits are indistinguishable in their own class:  $1 + 1 = 2$  ways.
3. **3** ways.
4. 2 B's, 2 G's, or 1 B and 1 G: **3** ways.
5. 6 students total (3 boys + 3 girls) and we choose 2:  $C(6, 2) = 15$  ways.
6. **1** way, since any orange we do not pick is indistinguishable from any other orange that we did not pick in a different scenario.
7.  $C(6, 5) = 6$  ways.
8.  $C(6, 1) = 6$  ways.
9. We need to pick exactly 5 fruits. Let's consider picking  $i$  oranges and  $(5 - i)$  apples where  $0 \leq i \leq 5$ :
  - 0 oranges, 5 apples
  - 1 orange, 4 apples
  - 2 oranges, 3 apples
  - 3 oranges, 2 apples
  - 4 oranges, 1 apple
  - 5 oranges, 0 apples

Total: **6** ways.

10. Counting the different ways to pick each fruit:

- For oranges: 0 to 9 (10 choices)
- For apples: 0 to 6 (7 choices)

Therefore the total choice combinations are  $10 \times 7 = 70$  ways. But we have to subtract the one case where we pick 0 of both fruits, so we have  $70 - 1 = 69$  ways.

## Permutation Formula

The number of ways to arrange  $r$  objects from  $n$  distinct objects (order matters) is given by:

$$P(n, r) = \frac{n!}{(n-r)!}$$

Also written as  ${}_nC_r$  or  $A(n, r)$ .

### Relationship between Permutation and Combination:

Since permutations consider order while combinations do not, we have:

$$P(n, r) = r! \times C(n, r)$$

This is because for each combination of  $r$  objects, there are  $r!$  ways to arrange them.

### Quickies - II

1. Multiplication principle: We pick 1 Latin book from 5 and 1 Greek book from 7:  $5 \times 7 = 35$  ways.
2. Each letter can be any of the 26 letters:  $26^2$  ways.
3. Since we can't repeat letters, we have 26 choices for the first letter and 25 for the second:  $26 \times 25 = 650$  ways.
4.  $21 \times 5 = 105$  ways.
5.  $3 \times 8 = 24$  ways.
6.  $P(5, 2) = 20$  ways. (We permute here since the arrangement matters)
7.  $C(5, 2) = 10$  ways.
8.  $26^4$  ways.
9. Pick any row (5 choices) and any column (7 choices):  $5 \times 7 = 35$  ways.
10.  $m \times n$  ways.
11. Coin has 2 outcomes, die has 6 outcomes:  $2 \times 6 = 12$  ways.
12.  $2 \times 6 \times 52 = 624$  ways.
13.  $4!$  ways (since each ace is distinct).
14.  $13!$  ways.

### A Discussion Question

**Question:** How many ways can a pair of dice fall?

Solution for this depends on how the question means, or how we interpret, "ways":

**Distinguishable Dice:** If we can tell the dice apart (e.g., one red die and one blue die), then each die can show any of 6 faces independently. Using the multiplication principle:  $6 \times 6 = 36$  ways.

This counts (1,2) and (2,1) as different outcomes since the first number represents the red die and the second represents the blue die.

**Indistinguishable Dice (Unordered Pairs):** If the dice are identical and we only care about which numbers appear, then we're counting unordered pairs. The possible outcomes are: (1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2), (2,3), (2,4), (2,5), (2,6), (3,3), (3,4), (3,5), (3,6), (4,4), (4,5), (4,6), (5,5), (5,6), (6,6)

This gives us  $\binom{6+2-1}{2} = \binom{7}{2} = 21$  ways (stars and bars approach).

**Possible Sums:** If we only care about the sum of the dice, there are 11 possible sums: 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

## Pigeonhole Principle

The pigeonhole principle is a fundamental counting principle:

If  $n$  objects are placed into  $m$  containers where  $n > m$ , then at least one container must contain more than one object.

This can be generalized to: if  $n$  objects are placed into  $m$  containers, then at least one container contains at least  $\lceil \frac{n}{m} \rceil$  objects.

## Pigeonhole Problems

1. First, draw a white sock. Then, draw a black sock. The next sock drawn will have to match one of these, thus giving us a pair. Therefore, we need **3** socks.
2. Same as last question, we have 4 different suits. We can draw one each from all of them, but the 5th drawn card will have to match one of the suits. Thus, we need **5** cards.
3. Given 365 possible birthdays, we can have two people for each birthdate ( $365 \times 2 = 730$ ), and one more person, who has to be born on one of these 365 days, resulting in 3 people having the same birthday. Therefore, we need **731** people.
4. We have 4 colors: 12 red, 20 white, 7 blue, 8 green balls. To avoid having 10 balls of the same color, we can take at most 9 from each color that has at least 9 balls. We can take all 7 blue balls, all 8 green balls, 9 red balls, and 9 white balls, giving us  $7 + 8 + 9 + 9 = 33$  balls. The 34th ball must be either red or white, giving us our 10th ball of that color. Therefore, we need **34** balls.
5. We can pick 1 person from each couple, giving us  $n$  people. The next person picked has to be the un-picked half of any couple, giving us  $n + 1$  people to ensure atleast one couple is picked.
6. Each person can have between 0 and 19 mutual friends (since there are 19 other people in the room). However, if one person has 0 mutual friends and another has 19 mutual friends, this creates a contradiction: if someone has 19 mutual friends, then everyone else is their friend, so no one can have 0 mutual friends. Therefore, there are only 19 possible values for the number of mutual friends. With 20 people and 19 possible values, by the pigeonhole principle, at least 2 people must have the same number of mutual friends. QED.
7. Consider any 5 lattice points. We can partition all lattice points into 4 classes based on the parity of their coordinates:
  - Class 1: (even, even)
  - Class 2: (even, odd)
  - Class 3: (odd, even)
  - Class 4: (odd, odd)

Two points are in the same class if and only if their x-coordinates have the same parity and their y-coordinates have the same parity. Since we have 5 points and only 4 classes, by the pigeonhole principle, at least 2 of the 5 points must be in the same class.

If two points  $(x_1, y_1)$  and  $(x_2, y_2)$  are in the same class, then their midpoint  $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$  has integer coordinates, since both  $x_1 + x_2$  and  $y_1 + y_2$  are even. This midpoint lies on the segment connecting the two points, so we have found a lattice point on one of our 10 segments. QED.

8. Label the 6 people as vertices of a complete graph  $K_6$ . Color each edge red if the corresponding people know each other, blue if they are strangers.

Pick any vertex  $v$ . It has 5 edges connecting to other vertices. By the pigeonhole principle, at least  $\lceil \frac{5}{2} \rceil = 3$  edges have the same color.

Without loss of generality, assume at least 3 edges from  $v$  are red, connecting  $v$  to vertices  $a$ ,  $b$ , and  $c$ . Now consider the triangle formed by  $a$ ,  $b$ , and  $c$ :

- If any edge of this triangle is red, then we have a red triangle (3 mutual acquaintances)
- If all edges of this triangle are blue, then we have a blue triangle (3 mutual strangers)

In either case, we have found the required set of 3. QED.

## Ramsey Theory

Ramsey theory studies the conditions under which order must appear in large enough structures. The fundamental question is: how large must a structure be to guarantee that it contains a particular substructure?

**Ramsey Number  $R(m, n)$ :** The smallest number  $N$  such that if we color the edges of the complete graph  $K_N$  with two colors (red and blue), then either there exists a red clique of size  $m$  or a blue clique of size  $n$ .

Problem 8 from the previous section demonstrates that  $R(3, 3) = 6$ . This means that in any group of 6 people, we can always find either 3 mutual acquaintances or 3 mutual strangers, and 6 is the smallest number for which this is guaranteed.

## Arrangements with Repetition

When arranging objects where some are identical, we must account for the fact that swapping identical objects doesn't create a new arrangement:

If we have  $n$  total objects consisting of  $n_1$  identical objects of type 1,  $n_2$  identical objects of type 2, ...,  $n_k$  identical objects of type  $k$ , then the number of distinct arrangements is:

$$\frac{n!}{n_1! \times n_2! \times \dots \times n_k!}$$

where  $n_1 + n_2 + \dots + n_k = n$ .

**Reasoning:** Start with  $n!$  total arrangements, as if all objects were distinct. However, since the  $n_1$  objects of type 1 can be arranged among themselves in  $n_1!$  ways without creating new distinct arrangements, we divide by  $n_1!$ . Apply the same logic for each type of identical object.

This formula naturally reduces to  $n!$  when all objects are distinct (each  $n_i = 1$ ) and to 1 when all objects are identical ( $n_1 = n$ , all other  $n_i = 0$ ).

## n Choose r by Way of MISSISSIPPI

1. All 6 letters are distinct, so we have  $6! = 720$  ways.
2. All 6 letters are distinct (subscripts make them different), so we have  $6! = 720$  ways.
3. 3 distinct A's, 2 identical E's, and 1 F.  $\frac{6!}{2!} = 360$  ways.

4. We have 3 identical A's, 2 distinct E's (due to subscripts), and 1 J. The number of arrangements is  $\frac{6!}{3!} = 120$  ways.
5. 3 identical A's, 2 identical E's, and 1 F.  $\frac{6!}{3! \times 2!} = 60$  ways.
6. 1 B, 3 A's, and 2 N's.  $\frac{6!}{1! \times 3! \times 2!} = 60$  ways.
7. 3 A's, 2 B's, 4 C's, and 1 D.  $\frac{10!}{3! \times 2! \times 4! \times 1!} = 12600$  ways.
8.  $\frac{11!}{2! \times 2! \times 2!}$  ways.
9.  $\frac{11!}{1! \times 4! \times 4! \times 2!}$  ways.
10. 4 A's, 3 G's, and 6 distinct letters (total 13 objects). The number of arrangements is  $\frac{13!}{4! \times 3!}$ .
11.  $\frac{13!}{4! \times 3!}$  ways.
12. First arrange the 3 subjects:  $3!$  ways. Then arrange books within each subject:  $4! \times 3! \times 6!$  ways. Total arrangements:  $3! \times 4! \times 3! \times 6! = 622080$ .
13. We need to arrange  $n$  letters where  $r$  are C's and  $(n - r)$  are R's. The number of arrangements is  $\frac{n!}{r!(n-r)!} = \binom{n}{r}$ .
14. Selecting  $r$  persons from  $n$  persons is  $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ .
15. Selecting  $r$  distinguishable objects from  $n$  distinguishable objects is  $\binom{n}{r} = \frac{n!}{r!(n-r)!}$  (This is the same as the last question, even though the objects are distinguishable. This is because the order of selection does not matter).

## Circular Arrangements

When arranging objects in a circle, we must account for the fact that rotations of the same arrangement are considered identical:

The number of ways to arrange  $n$  distinct objects in a circle is  $(n - 1)!$ .

**Reasoning:** Consider any linear arrangement of  $n$  people. When we place them in a circle, this single arrangement can be rotated  $n$  different ways around the circle, but all these rotations represent the same circular seating arrangement. Since there are  $n!$  linear arrangements, and each circular arrangement corresponds to  $n$  linear arrangements, we have  $\frac{n!}{n} = (n - 1)!$  distinct circular arrangements.

Equivalently, we can fix one person's position (to eliminate rotational symmetry) and arrange the remaining  $(n - 1)$  people in the remaining positions, giving us  $(n - 1)!$  arrangements.

## The Round Table

1. Using the circular arrangement formula:  $(8 - 1)! = 7! = 5040$  ways.
2.  $(12 - 1)! = 11!$  ways.
3. Treat each couple as a single unit. We have 8 units to arrange in a row:  $8!$  ways. Within each couple, the 2 persons can be arranged in  $2!$  ways. Total:  $2^8 \times 8!$  ways.
4. The couples can be arranged in  $(8 - 1)! = 7!$  ways, and within each couple there are  $2!$  arrangements. Total:  $2^8 \times 7!$  ways.
5. We have  $4 + 7 + 10 = 21$  people total. Using the circular arrangement formula:  $(21 - 1)! = 20!$  ways.

6. First, arrange the 8 R's in a circle:  $(8 - 1)! = 7!$  ways. This creates 8 gaps between consecutive R's where we can place the C's. To ensure no 2 C's are adjacent, we must choose 4 of these 8 gaps for our C's:  $\binom{8}{4}$  ways. Total:  $7! \times \binom{8}{4}$  ways.

## Homework

1.  $\binom{11}{5}$  ways.
2.  $\binom{52}{5}$  ways.
3.  $\binom{52}{13}$  ways.
4. A full house requires three-of-a-kind and a pair:
  - Choose rank for three-of-a-kind: **13** ways
  - Choose 3 cards from 4 of that rank:  $\binom{4}{3}$  ways
  - Choose different rank for pair: **12** ways
  - Choose 2 cards from 4 of that rank:  $\binom{4}{2}$  ways
 Total:  $13 \times \binom{4}{3} \times 12 \times \binom{4}{2}$  ways.
5.  $2^{10} - 1$  ways.
6.  $\frac{13!}{4! \times 4! \times 4! \times 1!}$  ways.
7. Total ways minus same-subject pairs:  $\binom{16}{2} - \binom{5}{2} - \binom{7}{2} - \binom{4}{2}$  ways.
8. Total combinations excluding choosing none:  $6 \times 8 - 1 = 47$  ways.
9. Arrange 21 consonants first: **21!** ways. This creates 22 gaps for the 5 vowels. Choose 5 gaps:  $\binom{22}{5}$  ways. Arrange vowels in chosen positions: **5!** ways.  
Total:  $21! \times \binom{22}{5} \times 5!$  ways.
10. First letter has 26 choices, each subsequent letter has 25 choices (cannot repeat previous):  **$26 \times 25^9$**  ways.
11. Use complement: total 10-element subsets minus those with no consecutive letters. For no consecutive letters, we choose 10 positions from an effective alphabet of size  **$26 - 9 = 17$** :  
 $\binom{26}{10} - \binom{17}{10}$  ways.
12.  $7! \times \binom{8}{5} \times 5!$  ways.
13.  $(7 - 1)! \times \binom{7}{5} \times 5!$  ways.

## Which of These Questions Can We Answer Now

1. **With repetition not allowed and order counting:** This is asking for permutations of  $r$  objects from  $n$  distinguishable objects. Answer:  $P(n, r) = \frac{n!}{(n-r)!}$  ways.
2. **With repetition allowed and order counting:** Each of the  $r$  positions can be filled with any of the  $n$  objects independently. Answer:  $n^r$  ways.
3. **With repetition not allowed and order not counting:** This is asking for combinations of  $r$  objects from  $n$  distinguishable objects. Answer:  $C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$  ways.
4. **With repetition allowed and order not counting:** This is combinations with repetition, solved using the stars and bars method. We need to distribute  $r$  identical items among  $n$  distinct categories. Answer:  $\binom{n+r-1}{r} = \binom{n+r-1}{n-1}$  ways.



## Combinations with Repetition

The number of ways to select  $r$  objects from  $n$  distinct types with repetition allowed is given by:

$$\binom{n+r-1}{r} = \binom{n+r-1}{n-1}$$

**Reasoning:** This problem is equivalent to distributing  $r$  identical objects among  $n$  distinct categories. We can visualize this using the “stars and bars” method: imagine  $r$  stars (representing our selections) and  $(n-1)$  bars (dividing the stars into  $n$  groups for each type). For example, selecting 5 items from types A, B, C might look like:  $\star\star|\star|\star\star$ , representing 2 A’s, 1 B, and 2 C’s.

We need to arrange  $r$  stars and  $(n-1)$  bars in a row, which gives us  $(r+n-1)$  total positions. We choose  $r$  positions for the stars (or equivalently,  $(n-1)$  positions for the bars), yielding  $\binom{n+r-1}{r}$  arrangements.

Equivalently, we can think of this as finding the number of non-negative integer solutions to  $x_1 + x_2 + \dots + x_n = r$ , where each  $x_i$  represents how many times we select type  $i$ .

### Problems for Class

1.  $\binom{6+k-1}{k}$  ways.
2.  $\binom{26+4-1}{4}$  ways.
3. Arrange 10 men first:  $10!$  ways. This creates 11 gaps for the 7 women. Choose 7 gaps:  $\binom{11}{7}$  ways. Arrange women in chosen positions:  $7!$  ways.  
Total:  $10! \times \binom{11}{7} \times 7!$  ways.
4. Fix one man to account for circular symmetry:  $(10-1)!$  ways to arrange remaining men. This creates 10 gaps around the circle. Choose 7 gaps:  $\binom{10}{7}$  ways. Arrange women:  $7!$  ways.  
Total:  $9! \times \binom{10}{7} \times 7!$  ways.
5. Letters: R-4, E-4, C-2, U-1, N-2, L-1, A-1, T-1, I-1, O-1 (8 vowels, 10 consonants).
  - Arrange consonants:  $\frac{10!}{4! \times 2! \times 2!}$  ways
  - This creates 11 gaps. Choose 8 for vowels:  $\binom{11}{8}$  ways
  - Arrange vowels in chosen positions:  $\frac{8!}{4!}$  ways
 Total:  $\frac{11!}{3! \times 2! \times 3! \times 1! \times 2!} \times \binom{12}{8} \times \frac{8!}{4!}$  ways.
6. Total arrangements divided by vowel arrangements, times 1 (alphabetical):  
 $\frac{19!}{3! \times 4! \times 2! \times 1! \times 3! \times 1! \times 1! \times 2! \times 1! \times 1!} \times \frac{4!}{8!}$  ways.
7. Total words minus those containing BAD:  $4^5 - 3 \times 4^2$  ways.
8. Total arrangements minus adjacent arrangements:  $8! - 7! \times 2!$  ways.
9. Treat Peter and Paul as one unit. Arrange 7 units in circle:  $6!$  ways. Arrange Peter and Paul within unit:  $2!$  ways.  
Total:  $6! \times 2!$  ways.
10. Each nationality forms 2 pairs. Arrange  $2n$  pairs:  $(2n)!$  ways. Divide by  $2^n$  since pairs within each nationality are indistinguishable. Assign 4 people to 4 positions per nationality:  $(4!)^n$  ways.  
Total:  $\frac{(2n)!}{2^n} \times (4!)^n$  ways.

11. Choose toys sequentially for each child:  $\binom{20}{4} \times \binom{16}{4} \times \binom{12}{4} \times \binom{8}{4} \times \binom{4}{4} = \frac{20!}{(4!)^5}$  ways.
12.  $\frac{18!}{5! \times 6! \times 7!}$  ways.
13. Groups are indistinguishable, so divide by  $3!$ :  $\frac{18!}{6!^3 \times 3!}$  ways.
14. Arrange 11 B's creates 12 gaps. Choose 7 gaps for R's:  $1 \times \binom{12}{7} \times 1$  ways.
15. MISSISSIPPI has M-1, I-4, S-4, P-2. Arrange non-I letters:  $\frac{7!}{1! \times 4! \times 2!}$  ways. This creates 8 gaps. Choose 4 for I's:  $\binom{8}{4}$  ways.  
Total:  $\frac{7!}{1! \times 4! \times 2!} \times \binom{8}{4}$  ways.
16.  $\binom{5+67-1}{5}$  ways.
17. Let Lucky get  $L = l + 7$  balls, others get  $A + E + F = 23$  with  $A + E \leq 20$ :  $\sum_{s=0}^{20} (s + 1)(24 - s)$  ways.
18. Find solutions to  $a + b + c = 18$  with  $0 \leq a \leq 7, 0 \leq b \leq 8, 0 \leq c \leq 9$  using inclusion-exclusion:

$$\binom{18+3-1}{3-1} - \binom{(18-8)+3-1}{3-1} - \binom{(18-9)+3-1}{3-1} - \binom{(18-10)+3-1}{3-1} + \binom{(18-8-9)+3-1}{3-1} + \binom{(18-8-10)+3-1}{3-1} = 28 \text{ ways.}$$

### Ten Problems for Homework

1.  $\binom{26+4-1}{4} = \binom{29}{4}$  ways.
2.  $\binom{26}{4}$  ways.
3. Choose 2 ranks for pairs:  $\binom{13}{2}$  ways. Choose 2 cards from each rank:  $\binom{4}{2}^2$  ways. Choose 1 rank from remaining 11:  $\binom{11}{1}$  ways. Choose 1 card from that rank:  $\binom{4}{1}$  ways.  
Total:  $\binom{13}{2} \times \binom{4}{2}^2 \times \binom{11}{1} \times \binom{4}{1}$  ways.
4. Choose vowel for middle position: **5** ways. Choose first letter from remaining 25: **25** ways. Choose third letter from remaining 24: **24** ways.  
Total: **5**  $\times$  **25**  $\times$  **24** ways.
5. Use complement: Total arrangements minus no adjacent I's. Total:  $\frac{11!}{1! \times 4! \times 4! \times 2!}$  ways. No adjacent I's: Arrange M,S,S,S,P,P first in  $\frac{7!}{4! \times 2!}$  ways, place 4 I's in 8 gaps:  $\binom{8}{4}$  ways.  
Answer:  $\frac{11!}{1! \times 4! \times 4! \times 2!} - \frac{7!}{4! \times 2!} \times \binom{8}{4}$  ways.
6.  $\binom{12+2-1}{2}$  ways.
7. Since groups of same size are indistinguishable:  $\frac{18!}{5! \times 5! \times 4! \times 4! \times 2! \times 2!}$  ways.
8. Arrange objects into groups then account for indistinguishable piles:  $\frac{(mn)!}{(n!)^m \times m!}$  ways.
9. Choose 0-5 oranges and 0-7 apples: **(5 + 1)(7 + 1)** ways.
10. Sum arrangements for all valid selections of A's and B's (at least 1 letter):  $\sum_{i=0}^3 \sum_{j=0}^3 \frac{(i+j)!}{i! \times j!} - 1$  ways.

### Five Problems for Homework

1. Case 1 (P's together): Arrange M,I,I,I,PP in  $\frac{6!}{4!}$  ways, then place 4 S's using stars and bars:  $\binom{(7-2)+4-1}{4}$  ways.  
Case 2 (P's separate): Arrange M,I,I,I,I in  $\frac{5!}{4!}$  ways, place P's in  $\binom{6}{2}$  ways, then place S's:  $\binom{(8-4)+4-1}{4}$  ways.

Total:  $\frac{6!}{4!} \times \binom{9}{4} + \frac{5!}{4!} \times \binom{6}{2} \times \binom{7}{4}$  ways.

2.  $\binom{6+32-1}{32}$  ways.

3. Substitute  $y_i = x_i - 1$  to get  $y_1 + y_2 + y_3 + y_4 + y_5 + y_6 = 26$  with  $y_i \geq 0$ :

$\binom{6+26-1}{26}$  ways.

4. Consider  $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 32$  with  $x_7 \geq 1$ . Substitute  $y_i = x_i - 1$  for all variables to get  $y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 = 25$  with  $y_i \geq 0$ :

$\binom{7+25-1}{25}$  ways.

5. Consider  $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 32$  with  $x_7 \geq 1$ . Substitute  $y_7 = x_7 - 1$  to get  $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + y_7 = 31$  with all variables nonnegative:

$\binom{7+31-1}{31}$  ways.

### Ice Cream Cones - The Double Dip

1. Total flavour combos available for one icecream are  $12 \times 12$ . After that it's simply a matter of choosing with repetition amongst these:

$\binom{12 \times 12 + 5 - 1}{5}$  ways.

2.  $\binom{12 \times 12}{5}$  ways.

3. We must find out the number of total possible scoop combinations, given that the order does not matter. Then, we choose 5 of these combos:

$\binom{\binom{12+2-1}{2}}{5}$  ways.

4. Similar to above, and the first part of the calculation follows exactly. We simply choose with repetition allowed once we have the flavour combos:

$\binom{\binom{12+2-1}{2} + 5 - 1}{5}$  ways.

### Block Walking